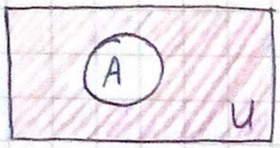
Задача 1.6

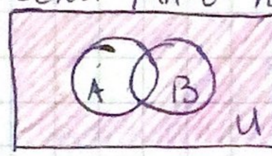
Диаграммы Венна

1.6.1

а) A' $\overline{A}$

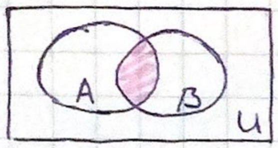


если мн-в несколько

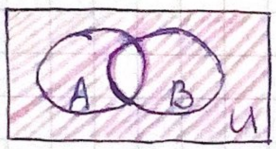


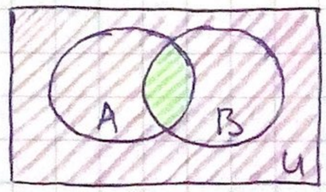
б) $\overline{A\cap B}$

$A\cap B$



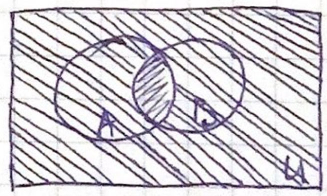
$\overline{A\cap B}$





$A\cap B$ $\overline{A\cap B}$

или



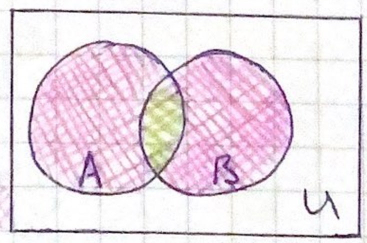
$A\cap B$ $\overline{A\cap B}$

в) $(A\cup B) - A\cap B$

1) $A\cup B$

2) $A\cap B$

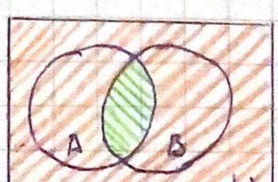
3) $(A\cup B) - A\cap B$



г) $A - \overline{B}$

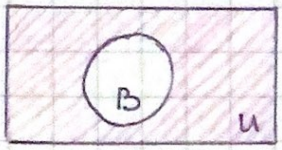
1) $\overline{B}$

2) $A - \overline{B}$



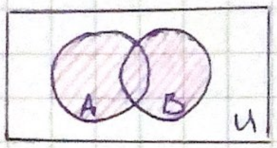
1.6.2

a) $\overline{B}$

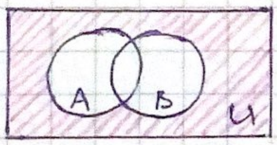


б) $\overline{A \cup B}$

1) $A \cup B$

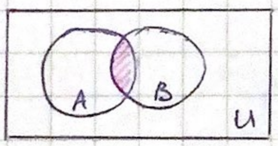


2) $\overline{A \cup B}$

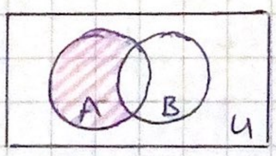


в) $A - A \cap B$

1) $A \cap B$



2) $A - A \cap B$



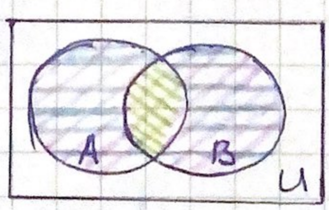
2) $A \triangle B =$

$= (A \cup B) \setminus (A \cap B)$

1) $A \cup B$

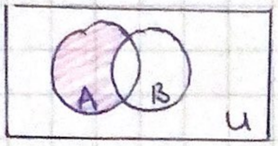
2) $A \cap B$

3) $(A \cup B) \setminus (A \cap B)$



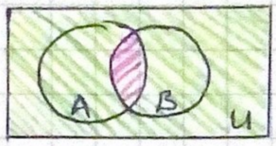
1.6.3.

а) $A - B$



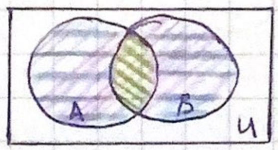
б) $(A \cap B)'$

1) $A \cap B$
2) $(A \cap B)'$



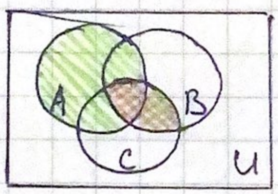
в) $(A \cup B) - A \cap B$

1) $A \cup B$
2) $A \cap B$
3) $(A \cup B) - A \cap B$



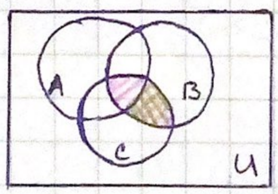
г) $A \cup (B \cap C)$

1) $B \cap C$
2) $A \cup (B \cap C)$



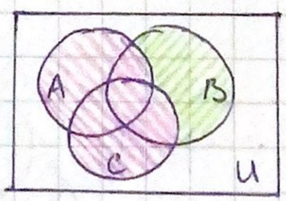
д) $(B \cap C) - A$

1) $B \cap C$
2) $(B \cap C) - A$



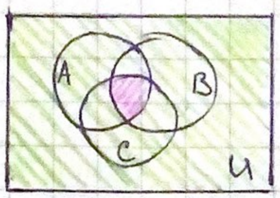
е) $B - (A \cup C)$

1) $A \cup C$
2) $B - (A \cup C)$



ж) $(A \cap B \cap C)'$

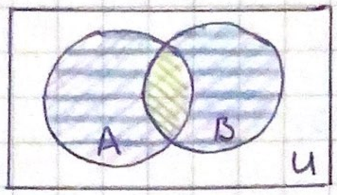
1) $A \cap B \cap C$
2) $(A \cap B \cap C)'$



1.6.4.

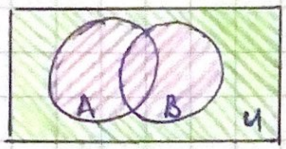
а) $A \triangle B = (A \cup B) \setminus (A \cap B)$

1) $A \cup B$
2) $A \cap B$
3) $(A \cup B) \setminus (A \cap B)$



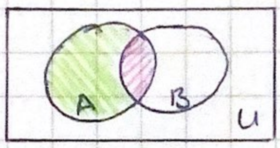
б) $(A \cup B)'$

1) $A \cup B$
2) $(A \cup B)'$



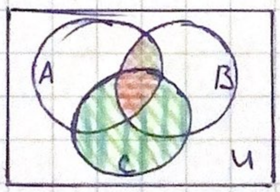
в) $A - A \cap B$

1) $A \cap B$
2) $A - A \cap B$



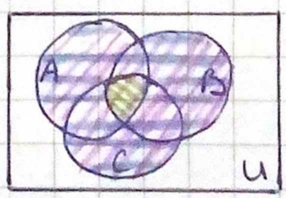
г) $(A \cap B) \triangle C$

1) $A \cap B$
2) $(A \cap B) \cup C$
3) $(A \cap B) \cap C$
4) $(A \cap B) \triangle C$



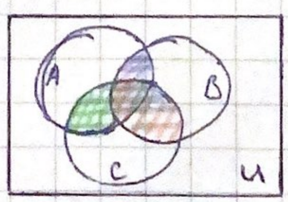
д) $(A \cup B \cup C) - (A \cap B \cap C)$

1) $A \cup B \cup C$
2) $A \cap B \cap C$
3) $(A \cup B \cup C) - (A \cap B \cap C)$



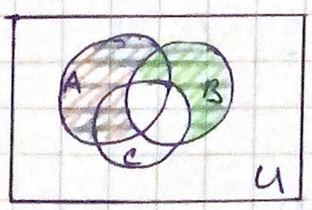
е) $(A \cap B) \cup (B \cap C) \cup (A \cap C)$

1) $A \cap B$
2) $B \cap C$
3) $A \cap C$
4) $(A \cap B) \cup (B \cap C) \cup (A \cap C)$



ж) $(A - B) \cup (B - C)$

1) $A - B$
2) $B - C$
3) $(A - B) \cup (B - C)$

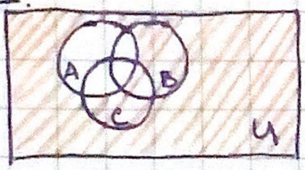


Задача 1.7.

Определить мн-во, соответствующ. закрашенной части диаграммы Венна

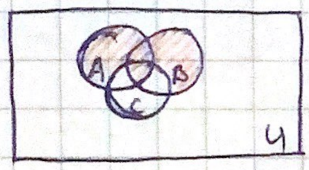
1.7.1.

а)



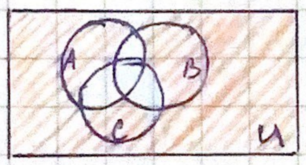
$\overline{A}$

б)



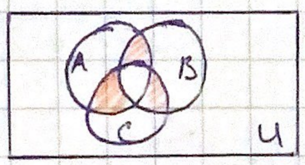
$(A\cup B-C)\cup\underbrace{(A\cap B\cap C)}_{A\cap B}$

в)



$((A\cap B)\cup(B\cap C)\cup(A\cap C))'$

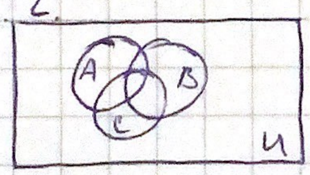
г)



$((A\cap B)\cup(B\cap C)\cup(A\cap C))-(A\cap B\cap C)$

1.7.2.

а)

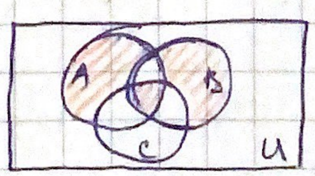


$\text{И}'$

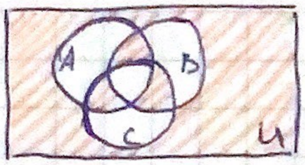
б)

$(A\cup B-C)\cup(A\cap B)$

в)



$(A-B\cup C)\cup(B-A\cup C)\cup(A\cap B\cap C)$

2) 

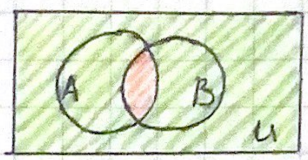
$$((A-B\cup C)\cup(B-A\cup C)\cup(C-A\cup B)\cup(A\cap B\cap C))'$$

Задача 1.8

показать $(A\cap B)'=A'\cup B'$ с помощью диаграммы Венна.

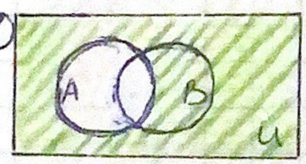
1) $(A\cap B)'$

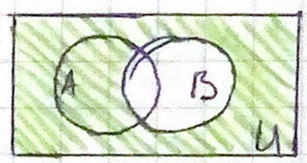
1. $A\cap B$
2. $(A\cap B)'$

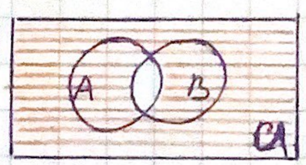


2) $A'\cup B'$

1. A'
2. B'
3. $A'\cup B'$

1 

2 

3. 

Итог $(A\cap B)'$ - зелёная штриховка в (1) и

итог $A'\cup B'$ - красная штриховка в (2)

совпадают $\Rightarrow (A\cap B)'=A'\cup B'$

Задача 1.9.

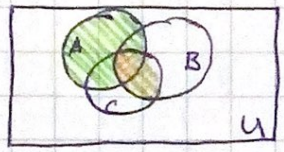
доказать $A\cup(B\cap C)=(A\cup B)\cap(A\cup C)$ с помощью диаграммы Венна

1) $A\cup(B\cap C)$

① $B\cap C$

② $A\cup(B\cap C)$

(1,2)



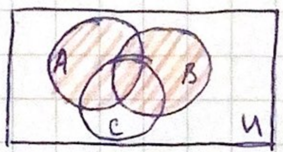
2) $(A\cup B)\cap(A\cup C)$

① $A\cup B$

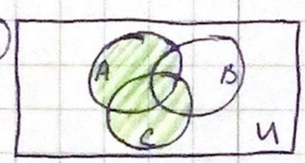
② $A\cup C$

③ $(A\cup B)\cap(A\cup C)$

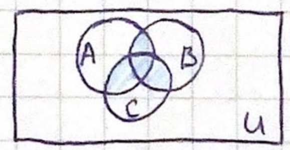
①



②



③



Задача 1.10.

Наследник мн-ва – это мн-во, состоящее из объединения элементов мн-ва A и самого мн-ва A

а) $\varnothing \to \varnothing\cup\{\varnothing\}$

б) $\{\varnothing\}\to\{\varnothing\}\cup\{\{\varnothing\}\}$

в) $\{\varnothing,\{\varnothing\}\}\to\{\varnothing,\{\varnothing\}\}\cup\{\{\varnothing,\{\varnothing\}\}\}$